Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 6 [2021 P2]

Test Date: 23rd Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2021-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 7 – 12)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 13 – 16)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 25)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 26 – 31)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 32 – 35)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 44)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 45 – 50)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 51 – 54)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Max. Marks: 60

SECTION-I
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

1. Two plane electromagnetic waves with the same amplitude E_0 and frequency ω travel in opposite directions along the z-axis, creating a standing wave. The electric field vectors of these waves are given by :

$\overrightarrow{E_1}(z,t) = E_0 \cos(kz - \omega t)\hat{i}, \overrightarrow{E_2}(z,t) = E_0 \cos(kz + \omega t + \phi)\hat{i}$

The superposition of these two waves forms a standing wave. Select the correct statements.

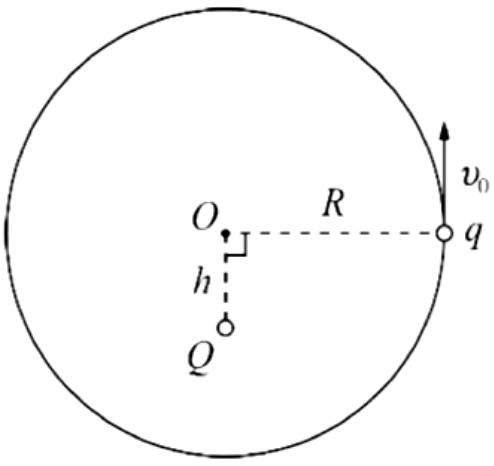
A) When $\phi = \frac{\pi}{2}$, the resulting wave is elliptically polarized and the polarization changes with position along the z-axis.

B) If $\phi = \pi$, the resulting wave is linearly polarized along a fixed direction in the xy-plane.

C) The magnetic field vector of the resulting wave oscillates perpendicular to the plane of polarization at each point.

D) For $\phi = \pi$, nodes of the electric field occur at positions $z = \frac{n\pi}{k}$, where n is an integer.

2. A small bead with mass m and charge +q can slide without friction along a non-conducting ring with radius R. In the plane of the ring, at a distance h from the center of the ring (h < R), a charge +Q is fixed. Initially (see figure), the velocity of the bead is v_0.



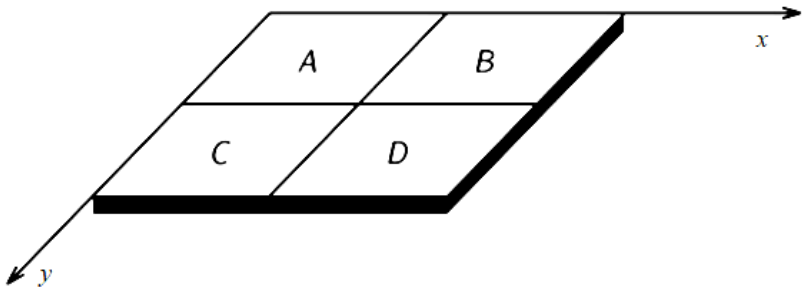
A) The maximum speed reached by the bead during the motion is $\sqrt{v_0^2 + \frac{2kqQ}{m}\left(\frac{1}{\sqrt{R^2+h^2}} - \frac{1}{R+h}\right)}$

B) The normal force acting on the bead when the speed is maximum, is zero.

C) The initial velocity required to complete a full revolution around the ring is $\sqrt{\frac{2kqQ}{m}\left(\frac{1}{R-h}-\frac{1}{\sqrt{R^2+h^2}}\right)}$.

D) The initial velocity required to complete a full revolution around the ring is $\sqrt{\frac{2kqQ}{m}\left(\frac{1}{R-h}+\frac{1}{\sqrt{R^2+h^2}}\right)}$.

3. A thin homogeneous square plate with side length a and mass m has its center of mass at its geometric center. The plate is divided into four equal parts A,B,C,D _ See Fig.



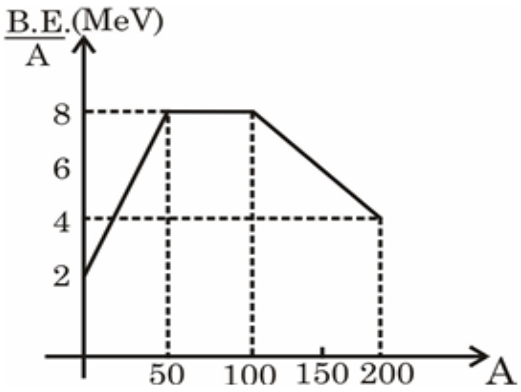
A) The co-ordinates of center of mass of system after removing D is $\left(\frac{5a}{12},\frac{a}{12}\right)$.

B) The co-ordinates of center of mass of system after removing D and placing it on part A is $\left(\frac{3a}{8},\frac{a}{8}\right)$.

C) The co-ordinates of center of mass of system after removing D and placing it on part A is $\left(\frac{3a}{8},\frac{3a}{8}\right)$.

D) The co-ordinates of center of mass of system after removing D and placing it on part B is $\left(\frac{a}{2},\frac{3a}{8}\right)$.

4. If the graph of binding energy $\left(\frac{B.E}{A}\right)$ per nucleon versus mass number (A) looks like as shown in figure, then using the curve choose correct options :-



A) Fusion of two nuclei with mass number 30 and 45 into one nucleons of mass number 75 will release energy.

B) Fission of nuclei with mass number 80 into two nuclei of equal mass number will have Q-value 256 MeV.

C) Fission of nuclei with mass number 150 into two nuclei of equal number will release energy.

D) Fusion of two nuclei with mass number 10 and 20 into a nucleus of mass number 30 will release energy.

5. You have three tuning forks, A, B, and C. Fork B has a frequency of 441 Hz; when A and B are sounded together, a beat frequency of 3 Hz is heard. When B and C are sounded together, the beat frequency is 4 Hz.

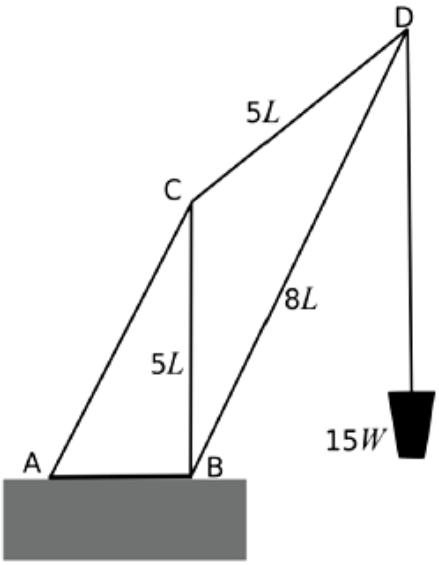
A) The frequency of A can be 438 Hz.

B) The frequency of C can be 444 Hz.

C) The beat frequency between A and C can be 1 Hz.

D) The beat frequency between A and C can be 6 Hz.

6. A winch is used to haul sand up from the bottom of a dry dock using a crane constructed from a very light framework of girders. The crane geometry can be seen in the diagram of figure. The crane is fixed to the horizontal dock side at points A and B. The dimensions of the sides CB, CD and BD are as shown, the girder CB is vertical, and the winch is hauling a load of weight 15W.



A) The force exerted on D by BD in 24W.

B) Tension in the rod CD is 15W.

C) If the force on the clamp at A by ground acts vertically downwards and has a magnitude 36W, the distance AB is 2L.

D) None of the above.

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 7 and 8

Question Stem

In the window of the physics laboratory, an inverted image of the neighboring school was photographed, created by a spherical glass flask with a radius of R = 60 mm filled with water. Take $\mu_{water} = \frac{4}{3}$.



7. At what distance d_1 (in cm) from the center of the flask is the center of this image located?

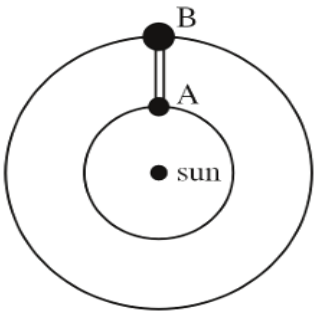
8. Since the camera is focused on the image of the building created by the flask, the actual school building is captured only as a blur. However, you can quite accurately compare the viewing angle under which you see the window of the building from the location where the center of the lens is located with the viewing angle under which you see, from the same location, the image of the window created by the flask. In the photograph, the height of the window appears as H = 50 mm, and the height of the image of the window created by the flask as h = 6 mm. What is the distance d_2 (in cm) from the center of the lens to the center of the image created by the flask ?

Question Stem for Question Nos. 9 and 10

Question Stem

Consider two planets, A and B, of masses $M_A = 1.0 \times 10^{24} kg$ and $M_B = 2.0 \times 10^{24} kg.$ in circular orbits of radius $R_A = 1.0 \times 10^{11} m$ and $R_B = 4 \times 10^{11} m$ around a common star of mass $M_S = 24 \times 10^{31} kg.$ The two planets are orbiting in the same direction. You may assume that the planets are point masses and have no rotation about their own individual axes.

$(Take : G = \frac{20}{3} \times 10^{-11} N - m^2 / kg^2)$



9. The planets have angular momentum L_A and L_B about the star. The value of $\frac{L_A}{L_B}$ is :

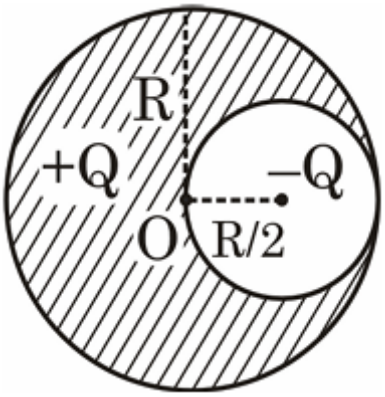
10. The inhabitants of the planets have constructed a light indestructible link bridge, which they connect when planets A and B are closest to each other. This joins the two planets instantaneously. Determine the angular velocity ω (in rad/year) of the two planets about their centre of mass at the instant just after the connection is made.

Question Stem for Question Nos. 11 and 12

Question Stem

Consider a non-conducting sphere of radius R with centre O. A spherical cavity of radius $\frac{R}{2}$ is created with it's centre $\frac{R}{2}$ from O. The remaining sphere has charge Q

uniformly distributed in it. The cavity is now filled with equal negative charge $-Q$, uniformly. $\left[Take\, Q = 14\mu C, R = 1cm\right]$



11. Find magnitude of potential energy of interation between positive and negative charges.

12. Find the net force on interaction between positive and negative charge (in kilo Newton)

SECTION-III
(PARAGRAPH WITH SINGLE CORRECT ANSWER TYPE)

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I

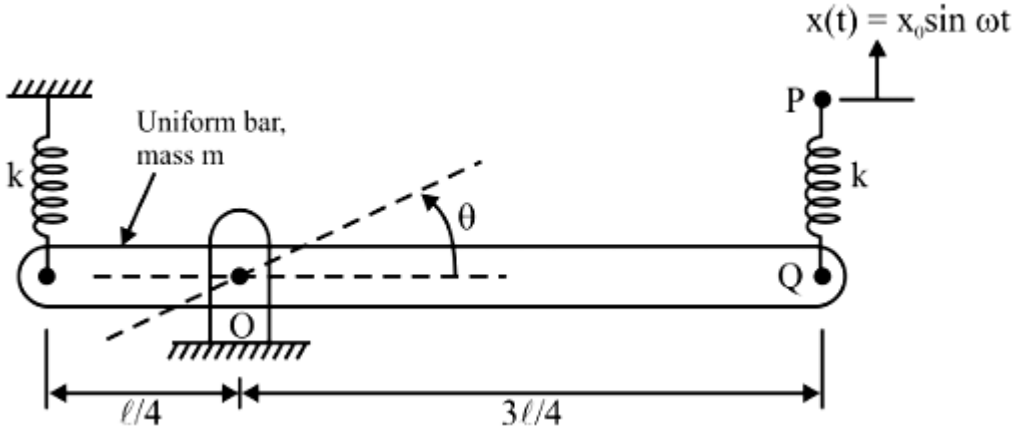
A uniform bar of mass m is pivoted at point O and supported at the ends by two springs, as shown in figure. End P of spring PQ is subjected to a sinusoidal displacement, $x(t) = x_0 \sin \omega t$.

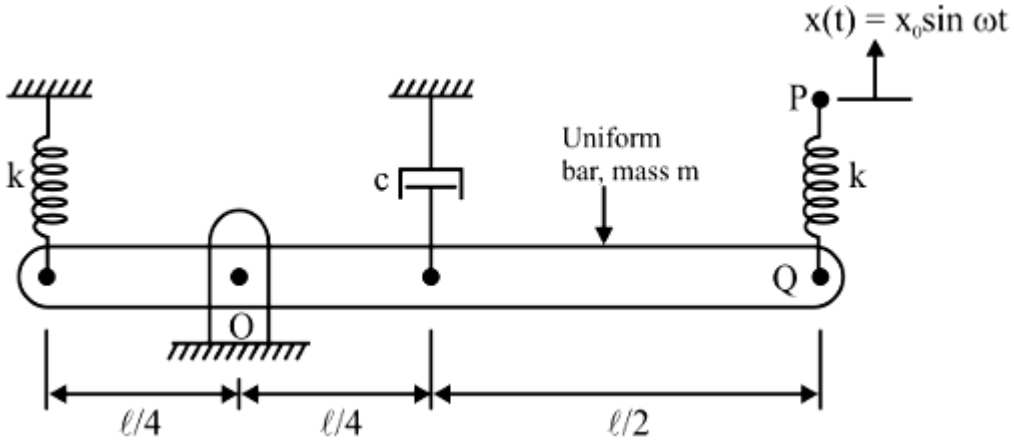
(ℓ = 1 m, k = 1000 N/m, m = 10 kg, $x_0 = 1cm$, and ω = 10 rad/s.)

Case-I : No damper is present.

Case-II : A damper having damping constant $C = \dfrac{1000\sqrt{2}}{3} N - s/m$ is attached.

(Neglect gravity)





13. The angular displacement of bar in case-I is

A) 0.01565 sin (10t) rad **B)** 0.01565 sin (10t + π) rad

C) 0.01565 sin $(10t + \frac{\pi}{2})$ rad **D)** 0.0313 sin (10t + π) rad

14. The angular amplitude of bar in case-II is

A) 0.008 rad **B)** 0.0131 rad **C)** 0.0262 rad **D)** 0.028 rad

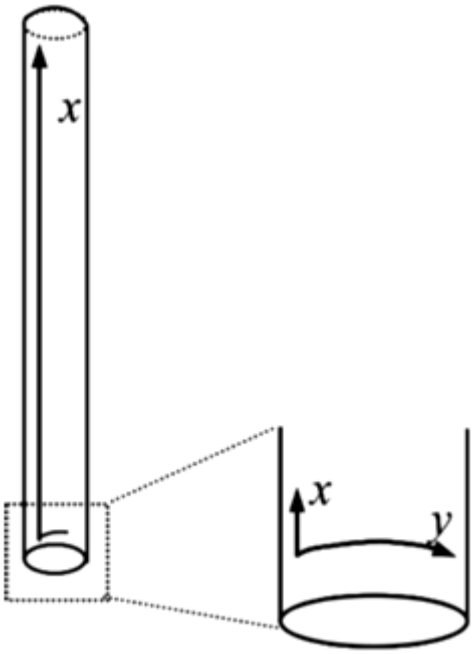
Paragraph-II

Consider a one-dimensional standing electromagnetic wave in the form of $E(x) = A\sin(k_x x)$ along the x-direction confined within the space between x = 0 and x = a. The wave must vanish at these two end points.

15. Find the allowed values of $k_x, (n = 1, 2, 3.....)$

A) $k_x = \frac{n\pi}{a}$ **B)** $k_x = \frac{2n\pi}{a}$ **C)** $k_x = \frac{n\pi}{2a}$ **D)** $k_x = \frac{4n\pi}{a}$

16 The String Theory predicts that our space is more than three-dimension, and the additional hidden dimensions are folded up like the dimension y on the surface of a thin cylinder shown in the figure. Suppose the radius of the cylinder is $b(<< a)$, and the electromagnetic wave on the surface now takes the form $E(x, y) = A\sin(k_x x)\cos(k_y y),$ where y is the coordinate of the folded space around the cylinder. Find the allowed values of k_y. $(m = 1, 2, 3.....)$



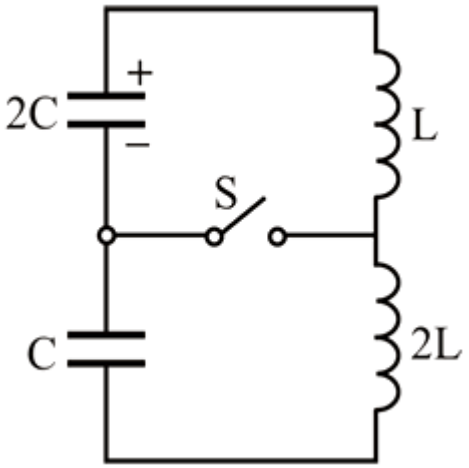
A) $k_y = \frac{m}{b}$ **B)** $k_y = \frac{m}{2b}$ **C)** $k_y = \frac{m}{\pi b}$ **D)** $k_y = \frac{2m}{\pi b}$

SECTION-4
Non-Negative Integer Answer Type

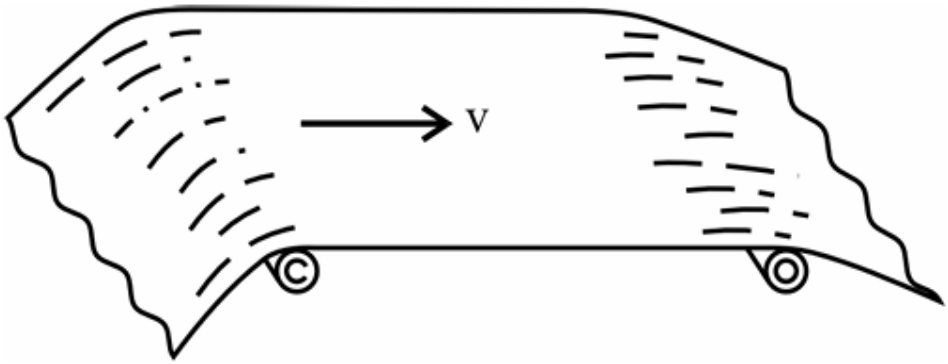
- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

17. Initially, a switch S open in the circuit shown in the figure, a capacitor of capacitance 2C carries the electric charge q_0 , a capacitor of capacitance C is uncharged, and there are no electric currents in both coils of inductance L and 2L, respectively. The capacitor starts to discharge and at the moment when the current in the coils reaches its maximum value, the switch S is instantly shorted. If the maximum current $I_{\max}$ through the switch S there after is $\frac{q_0}{\sqrt{\alpha LC}}$, the value of α is



18. In the production of polyethylene film, a wide strip is drawn over rollers at a speed of v = 45m/s as shown in the figure. During the processing (mainly due to friction), the surface of the film acquires a uniformly distributed charge. Find the maximum values of magnetic field induction B_{max} near the surface of the film, taking into account that at an electric field strength E = 20 kV/cm, an electrical discharge occurs in the air. The value of B_{max} is $\alpha \times 10^{-9} T$. The value of α is



19. A certain planet of radius R is composed of a uniform material that, through radioactive decay, generates a net power P. This results in a temperature difference between the inside and outside of the planet as heat is transferred from the interior to the surface. The thermal conductivity of the material is k. The temperature difference between the surface of the planet and the center of the planet is equal to $\frac{P}{\alpha \pi k R}$. The value of α is

CHEMISTRY

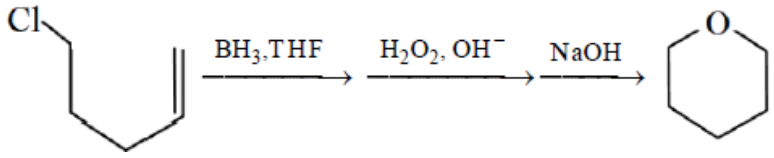
Max. Marks: 60

SECTION-I
(ONE OR MORE CORRECT ANSWER TYPE)

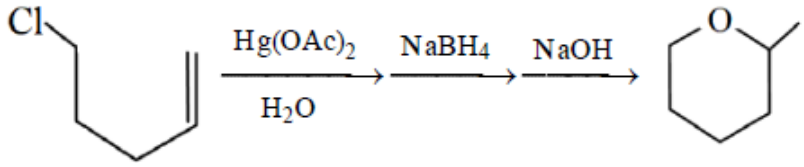
- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

20. Select the correct major products

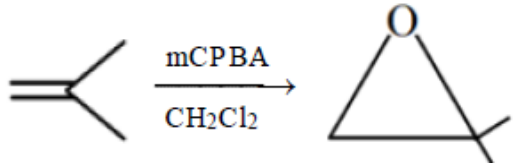
A)



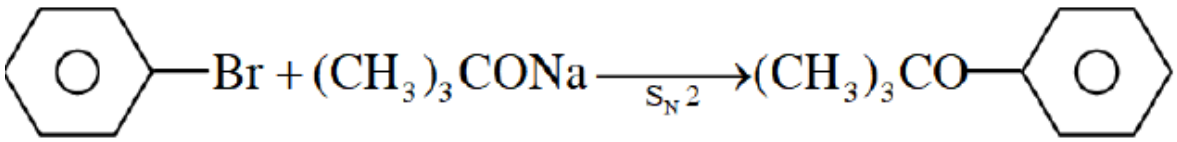
B)



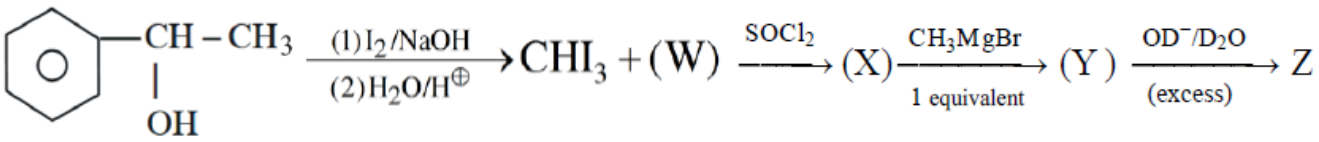
C)

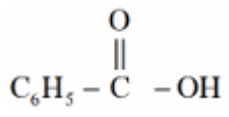


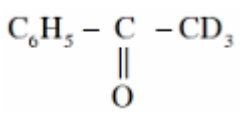
D)

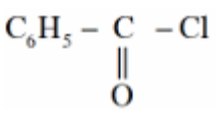


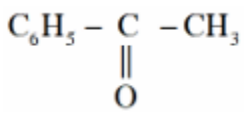
21. Which of the following option(s) have one of the products (s) obtained during this reaction sequence?



A) 

B) 

C) 

D) 

22. Select the correct option(s) regarding complex $K_2Zn_3[Fe(CN)_6]_2$:

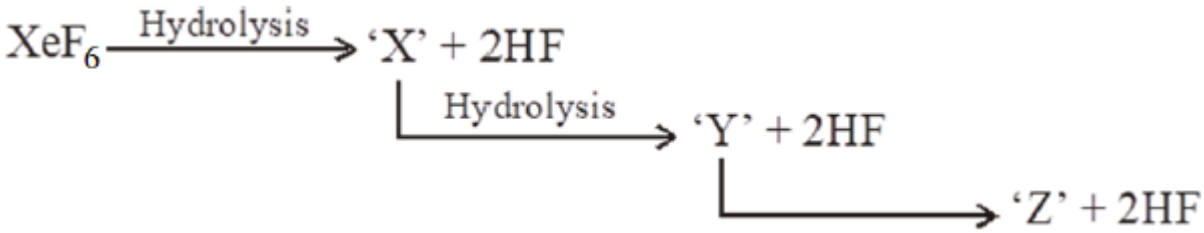
A) It is insoluble in water but dissolves in *exc.NaOH*.

B) Complex species is paramagnetic.

C) Central metal uses d^2sp^3 hybrid orbitals for bond formation.

D) Metal ligand linkage has $\pi - character$.

23. Select the correct statements (s)



A) 'Y' is see-saw shaped.

B) Central atom uses total three 'd-orbitals' for bonding in 'X'

C) 'Z' is planar but 'X' in non-planar

D) Z is white hygroscopic solid.

24. A metal cation M^{+2} gives black precipitate with acidified H_2S which dissolves in hot $conc.HNO_3$. Solution of M^{+2} gives brown precititate on treatment with Potassium ferrocyanide. Select the correct statement(s).

A) $M^{2+}(aq.)$ on reaction with $exc.KCN$ produces a gas

B) Metal 'M' in its extraction process from its most common sulphide ore is obtained from auto-reaction process

C) $E^0_{M^{+2}/M}$ is positive.

D) $M^{2+}(aq.)$ produce dark blue solution with excess Cl^-

25. P^H of which of the following solution in close to 5? (pka of $CH_3COOH = 5, K_a \ of \ NH_4^+ + 5.6\times10^{-10}$)

A) 50 mL of 0.01 M $CH_3COOH = 50mL \ of \ 0.01 \ M \ CH_3COONa$

B) 0.1 M solution of NH_4Cl

C) 0.1 M weak monoprotic acid having $K_a = 1\times10^{-9}$

D) $5\times10^{-6} M \ Ba(OH)_2$

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
  Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 26 and 27

Question Stem

An acid A which is an important constituent of vinegar, on reaction with red P and Br_2 gives a monobromoderivative (B) which on reaction with NH_3 gives white solid(C) . However ethanal on reaction with a mixture of ammonium chloride and sodium cyanide undergoes Strecker synthesis (synthesis of amino acids by the reaction of an aldehyde with cyanide in presence of ammonia) to give a product which on acidic hydrolysis gives another high melting solid(D). Let x and y are the total number of optical isomers of (C) and (D) respectively while z is the total number of dipeptides (excluding stereoisomers) that (C) and (D) can form.

26. The value of $x + y$ *is* ____

27. The value of z is____

Question Stem for Question Nos. 28 and 29

Question Stem

Consider the following sequences of reactions.

$(i)\ PbO_2 + conc.\ HNO_3 \rightarrow Pb(NO_3)_2 + H_2O + \underline{X}(g)$

$(ii)\ 2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + \underline{Y}(g)$

$(iii)\ FeS_{2(8)} + X(g) \rightarrow Fe_2O_3 + \underline{Z}(g)$

28. Assume ideal behavior of all three gases, 0.5,moles of each X and Y gas are taken in a container and expanded reversibly and adiabatically from $1\ dm^3$ *to* $4dm^3$ starting from $27^0 C$.Find final temperature(K).

Given $(4)^{0.4} = 1.74$

29. If equal moles of X, Y, Z (assuming ideal behavior of the gas) are expanded adiabatically and reversibly from the same initial state to same final volume. The magnitude of work is maximum for a gas. What is the value of γ for that gas, if all the degree of freedom are active.

Question Stem for Question Nos. 30 and 31

Question Stem

The following reactions are occurring in solvay process used for the manufacture of sodium carbonate.

$$(A) \xrightarrow{\Delta} (B) + CO_2$$

$$(B) + H_2O \longrightarrow (C)$$

$$(C) + NH_4Cl \rightarrow (D) + M$$

$$\xrightarrow{H_2O} \text{[D is a gas which is soluble in } H_2O\text{]}$$

$$(E)\ \text{solution}$$

$$(E) + CO_2(\text{excess}) \longrightarrow (F)$$

$$(F) + NaCl \longrightarrow (G) \xrightarrow{\Delta} \text{sodium carbonate.}$$

$\left(\begin{array}{l} Given\ that\ Atomic\ Mass: Na = 23,\ K = 39,\ Ca = 40,\ Mg = 24, \\ Fe = 56,\ Cu = 63.5,\ O = 16,\ S = 32,\ H = 1, C = 12,\ Cl = 35.5,\ F = 19,\ Zn = 65 \end{array}\right)$

30. If the molar mass of G *is* X $g\ mole^{-1}$ and that of (D) is Y $g\ mole^{-1}$. Then $X+Y$ *is* __

31. The number of molecules of F are obtained on reaction of $\left(\frac{1}{2}\right)$ mole of (E) with $\left(\frac{1}{4}\right)$ mole of CO_2 *is* $Q\times10^{21}$. What is the value of Q

SECTION-III
(PARAGRAPH WITH SINGLE CORRECT ANSWER TYPE)

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I:

1g of methane diffused in 20 seconds under certain conditions. Under the same conditions, $\sqrt{20g}$ of a open chain hydrocarbon (A) diffused in 40 seconds . A 10 mg of sample of (A) consumes 8.40 ml of H_2 gas measured at 0^0C and 760 mm pressure.

One of the open chain isomers of (A) on ozonolysis gives only formaldehyde & glyoxal.

32. A six membered stable monocyclic isomer of (A) on reductive ozonolysis can Not give which of the following products

A) $OHC-CHO$ **B)** $OHC-CH_2-CHO$

C) $OHC-CH_2-CH_2-CHO$ **D)** $OHC-CH_2-CH_2-CH_2-CHO$

33. Identify the incorrect statement.

A) One of the isomers of (A) exhibits geometrical isomerism

B) One of the isomers of (A) gives white precipitate with Tollen's reagent

C) One of the isomers of (A) is optically active.

D) One of the isomers of (A) on reductive ozonolysis gives only acetone.

Paragraph-II:

Consider the following series of reactions

$$\underset{2\ mole}{NaN_3(s)} \xrightarrow[T<300^0C]{\Delta} \underset{(solid)}{A} + \underset{(gas)}{B}$$

$$\underset{(gas)}{B} + \underset{excess}{H_2(g)} \rightarrow C(gas)$$

$$C(g) + bleaching\ \ powder\ \ solution \rightarrow B(g) + D + H_2O$$

34. Select the correct statements.

i) B is a diamagnetic gas having π bond order equal to 2.

ii) In molecular orbital diagram of (B) gas $H.O.M.O.$ contains 1 nodal plane.

iii) Compound C is a weak acid.

iv) Compound 'D' is obtained as byproduct of Solvay process.

A) Only (iv) **B)** (i), (iii) and (iv)

C) (i) and (iv) **D)** (i) ,(ii) and (iii)

35. Compound 'D' can be used to dry:.

A) NH_3 **B)** C_2H_5OH

C) Both (A) and (B) **D)** Neither 'A' nor 'B'

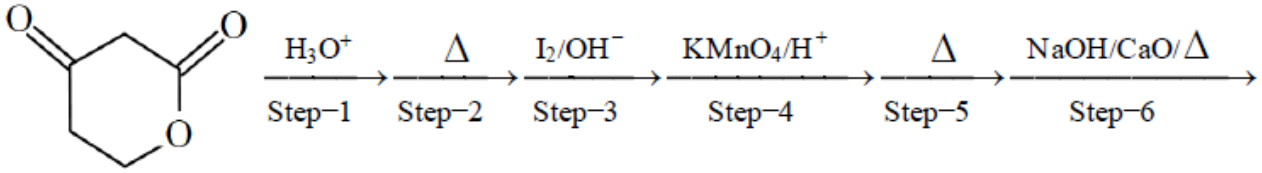
SECTION-4

Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

36. In how many steps decarboxylation reaction is taking place.



37. Two mole of non-volatile weak electrolyte XY was dissolved in 20 dm^3 of water . If boiling point of the solution was found to be 100.08^0C and K_b *of* H_2O *is* 0.5, then calculate the conductivity of the solution in $S\,m^{-1}$ unit. Assuming molality is equal to molarity. Λ_m^0 of $XY = 500 Scm^2$ / mole

38. $200\ cm^3$, 0.2 $M\ BaCl_2(aq.)$ is mixed with $0.1M, 600\ cm^3\ Na_2SO_4$ solution . Find osmotic pressure of resulting solution (in atm). Round off up to nearest integer value.

$R = 0.0821 L - atm\ K^{-1} mol^{-1}$

$T = 37^0C$

MATHEMATICS

Max. Marks: 60

SECTION-I
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
Negative Marks : −2 In all other cases.

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39. Let f(x) be a polynomial function satisfying $0 < xf(y) < yf(x) \forall x, y$ such that 0< x<y<1 and f(0)=0 then :

A) $f'(x) < f(1)$ **B)** $f(1) < 2\int_0^1 f(x)$ **C)** $3f\left(\frac{1}{3}\right) > 2f\left(\frac{1}{2}\right)$ **D)** $6f\left(\frac{1}{6}\right) < 5f\left(\frac{1}{5}\right)$

40. Let a, b , c be distinct complex numbers with $|a|=|b|=|c|=1$ and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Also P and Q are the points representing the complex number z_1 and z_2 respectively in the complex plane with $\angle POQ = \theta$ (where O being the origin) then which of the following is/are correct?

A) $b^2 = ac$ **B)** $\theta = \frac{2\pi}{3}$ **C)** $PQ = \sqrt{3}$ **D)** $|z_1 + z_2| = 1$

41. Let $< T_n >$ be a sequence such that $T_n^3 + 2T_n = T_{n+1} \forall n \in N$, *and* $T_1 = 1$, and T = 1, then :

A) $\sum_{n=1}^{100}\left(T_n^3 + T_n + 1\right) = T_{101} + 99$ **B)** $\sum_{n=1}^{100}\left(T_n^3 + T_n + 1\right) = T_{101} + 100$

C) $\prod_{n=1}^{100}\left(T_n^2 + 2\right) = T_{100}$ **D)** $\prod_{n=1}^{100}\left(\frac{T_{n+1}}{T_n} - T_n^2\right) = 2^{100}$

42. Let straight line y = mx + 4 meets the curve $3x^2 - (1-3a)xy - ay^2 = 0$ at two points A and B such that $\angle AOB = 90^\circ \forall m \in R - \{m_1, m_2\}$ where $m_1 < m_2$ and 'O' is the origin. Identify which of the following statements is/are correct ?

A) $m_1 + m_2 = \frac{10}{3}$

B) $am_1 + m_2 = 2$

C) If m = 2, then area of $\Delta AOB = \frac{80}{7}$ sq. units.

D) If m = 2, then area of $\Delta AOB = \frac{85}{7}$ sq. units

43. Let $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ and $B = \begin{bmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ac-b^2 \end{bmatrix}$

be two non-singular matrices such that $\left(A^2 - 2I\right)B = 0$ where a>b>c>0, then which of the following statements is/are correct?

[**Note:** I is an identify matrix of order 3 and Tr.(P) and det.(P) denote trace and value of the determinant of square matrix P respectively.]

A) $\text{Tr.}(AB) = 6\sqrt{2}$ **B)** $\text{Tr.}(AB) = -6\sqrt{2}$

C) $\det.\left(A - \sqrt{2}B\right) = 54\sqrt{2}$ **D)** $\det.\left(A - \sqrt{2}B\right) = -54\sqrt{2}$

44. If $\lambda_1 > \lambda_2 > \lambda_3 > \ldots > \lambda_n$ be the values of 'λ' for which equation $x^3 - 3x + \lambda = 0$ has all integral roots, if $h(x) = x^3 - 3x + |\lambda_1| + |\lambda_2| .... + |\lambda_n|$ for $x \in [1, \infty)$ and g(x) be the inverse of h(x) then:

A) $(\lambda_1)^2 + (\lambda_2)^2 + ... + (\lambda_n)^2 = a,$ number which is not the perfect square of an integer.

B) Area bounded by g(x) and y-axis between limits y=2 to y=5 is $\frac{531}{4}$.

C) $\int_1^3 \left(x^3 - 3x\right)dx + \int_2^{22} g(x)dx = 64$

D) $\int_1^3 \left(x^3 - 3x\right)dx + \int_2^{22} g(x)dx = 56$

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 45 and 46

Question Stem

An ellipse whose major axis is parallel to x-axis such that segment of a focal chord are 1 and 2 units. The line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ are the chords of the ellipse such that a, c, b are in H.P. and are bisected by the point at which they are concurrent, the equation of auxiliary circle is $2x^2 + 2y^2 + 4px + 4qy - 16p + 1 = 0$ then

45. Find p + q.

46. If length of latus rectum of ellipse is L, then find [L]. (where [.] represent Greatest integer function)

Question Stem for Question Nos. 47 and 48

Question Stem

If n be a natural number define polynomial $f_n(x)$ of nth degree as follows

$f_n(\cos\theta) = \cos n\theta$ i.e. $f_2(x) = 2x^2 - 1$; $f_3(x) = 4x^3 - 3x$. Then.

47. $\left(x + \sqrt{x^2-1}\right)^{10} + \left(x - \sqrt{x^2-1}\right)^{10}$ is equal to $\lambda\, f_k(x)$. Then find value of $\lambda + k$.

48. $f_6(x)$ is equal to $px^6 + qx^4 + rx^2 + s$ then find value of $7(p + q + r - s)$

Question Stem for Question Nos. 49 and 50

Question Stem

Let $P_1 : x + y + 2z - 3 = 0$ and $P_2 : x - 2y + z = 4$ be two planes. Let A(1,3,4) and B(3,2,7) be two points in R^3.

49. The equation of plane P_3 through line of intersection of P_1 and P_2 and length of projection upon which of the line segment AB is greatest is equal to ax+by+cz+1=0, then find the value of a+b+c. (where $a, b, c \in I$)

50. The equation of plane P_4 through the line of intersection of P_1 and P_2 and the length of projection upon which of the line segment AB is least is equal to ax+by+cz–7=0, then find the value of a+2b+c. (where $a, b, c \in I$)

SECTION-III
(PARAGRAPH WITH SINGLE CORRECT ANSWER TYPE)

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I:

Consider, $f(x) = \lim_{n\to\infty} \dfrac{\text{sgn}\left(\sqrt{ac} - b\right)e^{nx} + x^2 + f}{2e^{nx+x} + x + d}$ where $a > b > c > 0$ and $d,\ f \in R$

[Note: sgn (y) denotes the signum function of y.]

51. If a, b and c are in A.P. and f (x) is continuous for all $x \in R$, then the value of (2f + d + 1) is equal to :

A) 0 **B)** 1 **C)** -1 **D)** $\dfrac{-1}{2}$

52. If a, b and c are in G.P. and f(x) is continuous for all $x \in R\,(d < 0)$, then number of solutions of the equation $f(x) = \left|\,|x - 4| - 2\right| - 1$ is (are) :

A) 0 **B)** 2 **C)** 3 **D)** 4

Paragraph-II:

Let B_n denotes the event that n fair dice are rolled once with $P(B_n)=\frac{1}{2^n}$, $n \in N$,

e.g. $P(B_1)=\frac{1}{2}$, $P(B_2)=\frac{1}{2^2}$, $P(B_3)=\frac{1}{2^3}$,.....$P(B_n)=\frac{1}{2^n}$. Hence $B_1, B_2, B_3.....B_n$ are pair wise mutually exclusive and exhaustive events as $n \to \infty$. The events A occurs with atleast one of the event $B_1, B_2, B_3.....B_n$ and denotes that the sum of the numbers appearing on the dice is S.

53. If even number of dice has been rolled, the probability that S = 4, is

A) very closed to $\frac{1}{2}$ **B)** very closed to $\frac{1}{4}$

C) very closed to $\frac{1}{8}$ **D)** very closed to $\frac{1}{16}$

54. Probability that greatest number on the dice is 4 if three dice are known to have been rolled, is

A) $\frac{37}{216}$ **B)** $\frac{64}{216}$ **C)** $\frac{27}{216}$ **D)** $\frac{31}{216}$

SECTION-IV
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

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55. The sum of infinite series

$\frac{1}{15}+\frac{1}{30}+\frac{1}{50}+\frac{1}{75}+.....\infty$ is $\frac{k}{20,}$ then the value of k is

56. Given that for $a, b, c, d \in R$, if $a\sec(200°)-c\tan(200°)=d$ and

$b\sec(200°)+d\tan(200°)=c$ then $\frac{a^2+b^2+c^2+d^2}{ac-bd}=p\cos ec(200^o)$ find the value of

p_____.

57. If $I_n=\int_0^\infty e^{-x}(\sin x)^n dx (n>1)$, then the value of $\frac{101I_{10}}{I_8}$ is equal to _____.